

### UNIT 1: Introduction to Compilers

#### Introduction

A compiler is a software tool that translates high-level programming language code into machine code. It plays a crucial role in program execution.

This unit introduces compiler structure and phases.

#### Detailed Explanation

##### Phases of Compiler

The main phases include lexical analysis, syntax analysis, semantic analysis, optimization, and code generation.

Each phase transforms the program step by step.

##### Compiler vs Interpreter

A compiler translates the entire program at once, while an interpreter executes line by line.

#### Applications

Compilers are used in programming languages and software development.

#### Summary

This unit introduces compiler fundamentals.

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### UNIT 2: Lexical Analysis

#### Introduction

Lexical analysis is the first phase of a compiler.

#### Detailed Explanation

##### Tokens and Lexemes

Tokens represent basic units of code, while lexemes are actual sequences.

##### Role of Lexer

It removes whitespace and identifies tokens.

#### Applications

Used in compilers and interpreters.

#### Summary

This unit explains lexical analysis.

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## **UNIT 3: Syntax Analysis**

### **Introduction**

Syntax analysis checks the grammatical structure.

### **Detailed Explanation**

#### **Parsing Techniques**

Top-down and bottom-up parsing.

#### **Parse Trees**

Represent program structure.

### **Applications**

Used in compilers.

### **Summary**

This unit explains syntax analysis.

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## **UNIT 4: Semantic Analysis**

### **Introduction**

Ensures program meaning is correct.

### **Detailed Explanation**

#### **Type Checking**

Ensures compatibility.

#### **Symbol Table**

Stores variable information.

### **Applications**

Used in compilers.

### **Summary**

This unit explains semantic analysis.

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## **UNIT 5: Code Generation**

### **Introduction**

Final phase generating machine code.

### **Detailed Explanation**

**Intermediate Code**

Represents program.

**Optimization**

Improves efficiency.

**Applications**

Used in compilers.

**Summary**

This unit explains code generation.